wanrongz@usc.edu
https://ZoeyZheng0.github.io/

Wanrong (Zoey) Zheng

(213) 706-3365
Orchard Ave, Los Angeles

Education

University of Southern California, Ph.D. student in Computer Science, Advisor: Prof. Laurent Itti — *May. 2024 — Present*
University of Southern California, M.S. in Computer Science, Advisor: Prof. Ram Nevatia & Prof. Laurent Itti — *Aug. 2020 — May. 2023*
South University of Science and Technology, Exchange in Computer Science — *Jan. 2021 — Jun. 2021*

RESEARCH INTERESTS

Vison-and-Language Navigation [2]: Developing zero-shot language-guided agents navigating in complex visual environments.
Multi-Modal Reasoning [1, 3, 4, 5, 6, 7]: Exploring the synergistic effects of diverse data for enhanced machine understanding.

Publications

1. **Wanrong Zheng***, Yunhao Ge*, Xingrui Wang, Di Wu, Yao Xiao, Xu Zhi, Linwei Li, Ziyan Wu, and Laurent Itti. **A Graphical Framework for Knowledge Exchange between Humans and Neural Networks**. *Under review.* [paper].
2. **Wanrong Zheng**, Yunhao Ge, and Laurent Itti. **Three-Step Nav: A Hierarchical Global–Local Planner for Zero-Shot Vision-and-Language Navigation**. *Annual Conference on Artificial Intelligence and Statistics (AISTATS'26).* [paper][supp].
3. **Wanrong Zheng***, Haidong Zhu*, Zhaoheng Zheng, and Ram Nevatia. **GaitSTR: Gait Recognition with Sequential Two-stream Refinement**. *IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM'24).* [paper][code].
4. Haidong Zhu, **Wanrong Zheng**, Zhaoheng Zheng, and Ram Nevatia. **ShARc: Shape and Appearance Recognition for Person Identification In-the-wild**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV'24).* [paper][slides].
5. Haidong Zhu*, **Wanrong Zheng***, Zhaoheng Zheng, and Ram Nevatia. **GaitRef: Gait Recognition with Refined Sequential Skeletons**. *IEEE International Joint Conference on Biometrics (IJCB'23),* **(Oral)**. [paper][code][project].
6. Haidong Zhu, Zhaoheng Zheng, **Wanrong Zheng**, and Ram Nevatia. **CAT-NeRF: Constancy-Aware Tx^{2}Former for Dynamic Body Modeling**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW'23).* [paper][code][supp].
7. Xiaoke Jiang, Yu Qiao, Junjie Yan, Qichen Li, **Wanrong Zheng**, and Dapeng Chen. **SSN3D: Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI'21).* [paper][supp][slides].

Employment

University of Southern California — **Los Angeles, CA**
Research Assistant, Advisor: Prof. Laurent Itti — *May. 2024 — Present*
Research Assistant, Advisor: Prof. Ram Nevatia — *Jan. 2022 — May. 2024*

SenseTime Technology Co., Ltd. — **Shenzhen, China**
Research Engineer, Advisor: Dr. Xiaoke Jiang & Dr. Yichao Wu — *Sep. 2019 — Aug. 2021*

Chinese Academy of Science, Shenzhen Institutes of Advanced Technology — **Shenzhen, China**
Research Assistant, Advisor: Dr. Qiong Wang — *Jul. 2018 — Jun. 2019*

The Chinese University of Hong Kong, Shenzhen Research Institute — **Shenzhen, China**
Research Intern — *Dec. 2017 — Jun. 2018*

Awards & Honors

1st on MS1M dataset in Masked Face Recognition Challenge (ICCV 2021) out of 136 teams — *Oct. 2021*
2nd on Glint360k dataset in Masked Face Recognition Challenge (ICCV 2021) out of 86 teams — *Oct. 2021*
National Endeavor Scholarship for Top Undergraduate Students of China (top 1%) — *Nov. 2017*

Service

Reviewer: CVPR 2026, ECCV 2026, AISTATS 2026, IEEE Transactions on Multimedia, IEEE Transactions on Cognitive and Developmental Systems, International Journal of Computer Vision

Research Experience

iLab, University of Southern California **Los Angeles, CA**
Research Assistant, Advisor: Prof. Laurent Itti *May. 2024 — Present*

- **Three-Step Nav: A Hierarchical Global–Local Planner for Zero-Shot Vision-and-Language Navigation**
  - Proposed a hierarchical global-local framework that alternates between *looking forward* to sketch global plans, *looking now* for fine-grained local grounding, and *looking backward* to audit execution history.
  - Designed an adaptive judge module equipped with meta-skills to detect navigation drift and trigger self-correction without task-specific fine-tuning.
  - Achieved state-of-the-art zero-shot performance on R2R-CE and RxR-CE datasets, reducing navigation error by 15% and improving SPL by 12% on R2R-CE validation splits.
  - One first-author paper accepted at AISTATS 2026 [2].
- **A Graphical Framework for Knowledge Exchange between Humans and Neural Networks**
  - Proposed a pipeline for humans to directly interact with Neural Networks on a structural representation of visual concepts.
  - Constructed Structural Concept Graphs (SCG), a reasoning logic mechanism of Neural Networks in classification tasks, using reasonable concepts extractor and Graph reasoning Network.
  - Humans could make decisions on the SCG and use SCG to guide the original Neural Network backward by knowledge distillation.
  - Accuracy increased by about 4% improvement on target ImageNet classes without a drop on the other classes.
  - Submitted one primary-author paper [1].

IRIS Computer Vision Lab, University of Southern California **Los Angeles, CA**
Research Assistant, Advisor: Prof. Ram Nevatia *Jan. 2022 — May. 2024*

- **ShARc: Shape and Appearance Recognition for Person Identification In-the-wild**
  - Developed a multimodal framework that integrates the pose and shape encoder with the aggregate appearance encoder for robust person identification in uncontrolled environments.
  - Achieved state-of-the-art results on public cloth-changing person re-identification datasets such as CCVID, MEVID, and BRIAR.
  - Published one co-author paper on WACV 2024 [4].
- **GaitRef: Gait Recognition with Refined Sequential Skeletons Knowledge Exchange**
  - Combined the silhouettes and skeletons information and refined the framewise joint predictions for gait recognition.
  - On Gait3D, the proposed method outperformed the baseline by 6.1% on Rank-1 and 5.4% on Rank-5.
  - Published one primary-author paper on IJCB 2023 (oral) [5].

Identity Verification, SenseTime **Shenzhen, China**
Research Engineer, Advisor: Dr. Yichao Wu & Dr. Ding Liang *Jan. 2021 — Aug. 2021*

- **Large-scale Phone Unlock Facial Verification**
  - Updated face unlock models for Chinese mobile phone manufacturers such as Huawei, Oppo, and Vivo.
  - Prepared three different size levels of models for various products' performance needs and used different training strategies.
  - Achieved 1e-6FAR@recall 90% for different races, including Caucasian, African, Asian, Indian, and Latino.

Smart City Group, SenseTime **Shenzhen, China**
Research Engineer, Advisor: Dr. Xiaoke Jiang & Dr. Junjie Yan *Dec. 2019 — Dec. 2020*

- **Self-Separated Network to Align Parts for 3D Convolution in Video Person Re-Identification**
  - Trained the Self-Separated Network in supervised / semi-supervised / unsupervised ways, which proved the efficiency of the semi-supervised alignment strategies, which used the labels with the selected position.
  - Received a 15.5% Rank-1 improvement on iLIDS compared to the fully supervised way.
  - Published one co-author paper on AAAI 2021 [7].

Patents

1. **Wanrong Zheng**, and Xiaoke Jiang. **A Identification method Based on History Passenger Flow Big Data.** *CN113269129A* (2021)
2. Sun Zhe, **Wanrong Zheng**, Xiaoke Jiang, Xinghua Yao, and Cong Ji. **Passenger Illegal Handing Bags Across Railing Detection in Real Railway Scene.** *CN112818844A* (2021)
3. **Wanrong Zheng**, Xiaoke Jiang, Jikui Bao, Qichen Li, and Cong Ji. **A Railway Face Recognition Solution Based on History Passengers' Riding Pattern.** *CN112232424A* (2020)